Sahyog Care Companion · Executive memo

Case study · Forward-deployed engagement for Sahyog, a two-city hospital network · 5 November 2026

The result

Sahyog has a working, independently reviewed assistant for all three of the board's mandates, tested on synthetic data. **It isn't ready for patients yet:** a supervised pilot can start once the three checks under "What happens next" are done. Patients book by describing symptoms in English, Hinglish or Hindi, after an emergency check. They ask questions about their own reports and get answers that name the page. Doctors sign a draft note made from their dictation. On synthetic data, the first build (the MVP, not re-run here) measured no patient's data in another patient's answer (0 in 26 attack attempts), a p95 triage turn of 3.3–4.3 seconds against a 4-second target, and about **$0.017 of model cost per consultation**, with infrastructure unpriced. What the board cares about most (calls avoided, time to book, and the doctor's time back) needs a pilot with real people to measure.

The problem

There are 9,000 appointments a day across Pune and Hyderabad. 41% are booked by phone, 18% of patients don't arrive, 2.4 million report PDFs are scanned a year, and doctors spend 11 minutes per consultation typing.

What we decided, and why

- **Code holds the hard lines.** Emergency rules run before any AI, and the patient filter is a database query. A prompt can be argued with, but a SQL filter can't.
- **A doctor signs every note.** A medicine, dose or "do not" the doctor didn't say becomes a question for the doctor, never an assumption.
- **Only scrubbed text leaves the hospital.** The patient's names and IDs are removed before any outside model call, and reports are read on site. Other people's names in free text can still get through.
- **Time blocks, not exact minutes.** A long visit is absorbed by its block, not passed on to every later patient.
- **Independent review of every change.** An AI reviewer that hadn't written the code checked each stage, and every finding was logged.

What review found

The reviews caught defects the first build's own tests had passed: a front-desk double booking, a relative's report filed under the wrong patient, emergency wordings missed in Hindi, and unsaid doses reaching a doctor's note. Each reported case was fixed with a test that would have caught it. One dose leak (a brand the model didn't list lending its dose) is still open. **The lesson:** an AI agent's tests check what it already thought of, so it needs a reviewer that didn't write the code.

What's not ready

There's been no independent emergency-recall test, no calibrated answer judge and no clinician review yet. Live-model results are from the first build. The word checks behind the refusal of diagnosis questions catch about one in four new phrasings. There are no backups and no real load test.

What we'd do differently

Protect the main branch so nothing merges before review passes: six of twelve changes merged early. Have people, not the build agent, write the held-out safety tests. Run with a model key from day one, so every number is this build's own.

What happens next

We recommended a four-week supervised pilot in one clinic, with front-desk staff watching every booking and doctors signing every note. Before it starts: a clinician's review of the rules and wording, emergency test phrases written by someone independent, and a privacy review of the model provider.